Graphs of Sine and Cosine Functions

Amplitude, period, and midline

1 For the function $y = 3\sin(2x)$, state:

a the amplitude **b** the period **c** the range

2 For the function $y = -2\cos\left(\frac{x}{2}\right) + 1$, state:

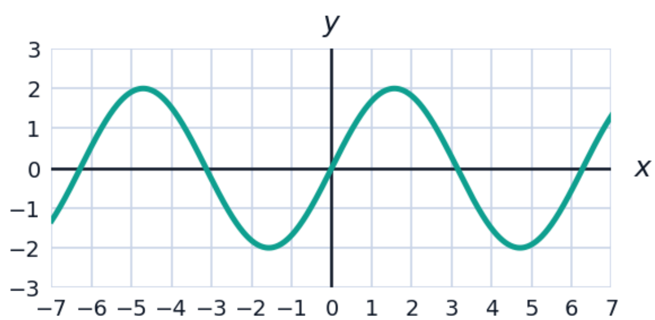
a the amplitude **b** the period

c the midline **d** the range

3 For $y = \sin\left(x - \frac{\pi}{3}\right)$, state the phase shift and the period.

Reading and writing equations from graphs

4 The graph below shows one sinusoidal function. State its amplitude and period, then write an equation for the function.



5 Write an equation of a cosine function with amplitude 4, period π, reflected across the x-axis, with no phase or vertical shift.

6 The depth of water at a harbour entrance is modeled by $d(t) = 2.5\sin\left(\frac{\pi}{6}t\right) + 7$ metres, where t is in hours after midnight.

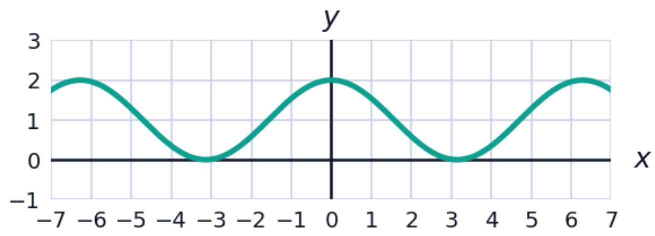
a State the maximum and minimum depths.

b State the period of the tide cycle.

c Find the depth at $t = 3$.

Graphing with transformations

7 The graph below shows $y = \cos(x) + 1$, a cosine curve with a vertical shift. Identify the midline and the maximum and minimum values shown.



8 Describe, in order, the transformations that turn $y = \sin x$ into $y = -3\sin\left(2\left(x - \frac{\pi}{4}\right)\right) + 2$.

9 State the amplitude, period, phase shift, and midline for $y = 5\cos\left(3x + \frac{\pi}{2}\right) - 4$. (Hint: factor the argument as $3\left(x + \frac{\pi}{6}\right)$.)

Modeling periodic phenomena

10 The average monthly temperature in a city varies between $8°$C (in January, month 1) and $28°$C (in July, month 7), following a sinusoidal pattern with a 12-month period.

a Find the amplitude and midline of the model.

b Write a cosine model $T(m)$ for the temperature in month m, using the fact that the minimum occurs at $m = 1$.

c Use your model to estimate the temperature in month 4 (April).